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Advanced Monitoring and Control Mechanism for LVDC Microgrid

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Abstract: An Advanced Monitoring and Control Mechanism for a 120V Low Voltage Direct Current (LVDC) Microgrid is presented in this project. The system uses an ESP-32 microcontroller to carry out MCU based algorithms and is made to manage dynamic loads and voltage variations. This control approach proactively maintains system stability in addition to optimizing power allocation between an Energy Storage System (ESS) and the 230V AC Grid-fed. The microgrid is extremely dependable and efficient due to load stabilization provided by a tight feedback control loop, which keeps the voltage and current levels on the DC bus constant.

Introduction:A significant development in contemporary power distribution is the shift to low voltage direct current microgrids, which provide high efficiency, lower conversion losses, and easier integration of energy storage and renewable energy sources. The main challenges in practical implementation, however, are preserving a steady voltage on the DC bus and maximizing power flow in the face of frequently fluctuating load conditions. Conventional, reactive control techniques frequently result in inadequate energy use and voltage drops. An advanced monitoring and control mechanism is the clever solution that this project suggests. By employing the potent ESP-32 to anticipate system requirements and proactively control the distribution of power from the Grid and the Energy Storage System, our system goes beyond simple reaction. The goal of this work is to create a very dependable and effective foundation for DC power infrastructure of the future.

Working: Sensing, prediction, and actuation are used to control the system's operation. In addition to the instantaneous voltage and current on the 120V DC Bus, several sensors continuously check the output from the backup system and the incoming power quality from the 230V AC Grid. The Microcontroller based algorithm on the ESP-32 uses this real-time data to determine the best control strategy for the upcoming time ahead. This forecast foresees continues load requirements and guarantees effective power usage. At the same time, load stabilization is provided by a fast-acting feedback control loop. The ESP-32 proactively mitigates the voltage drops and ensures a continuous, stable supply to all loads by immediately commanding the Change Over Relay and deploying the ESS to inject additional power when the bus line voltage drops as a result of unexpected demand.

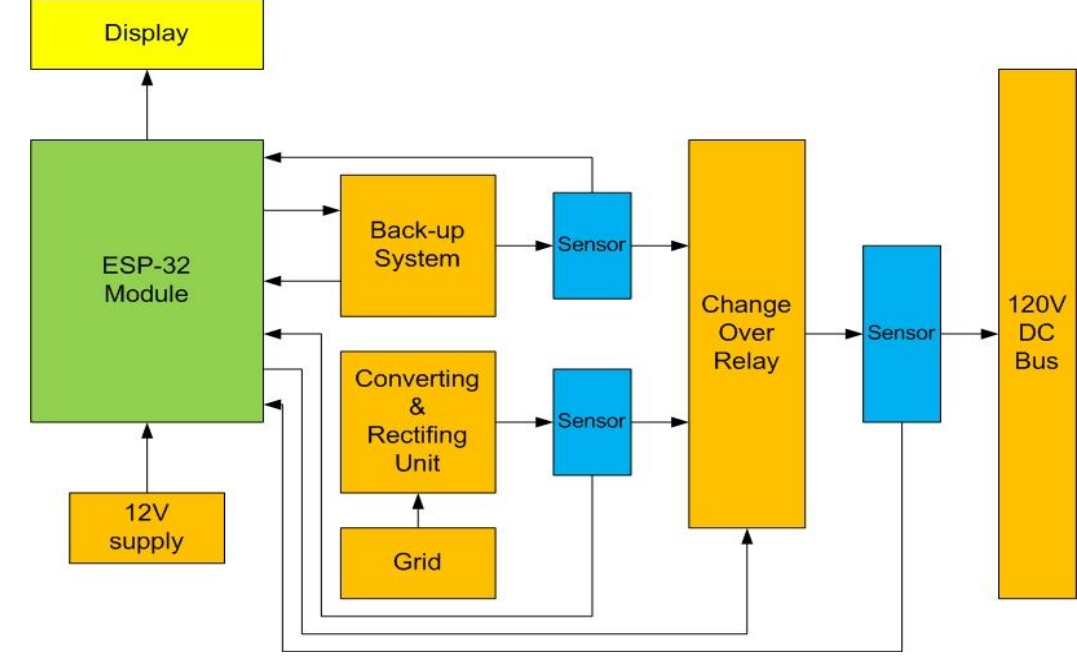
Methodology:A high-performance ESP-32 module is at the heart of the control architecture. Traditional sensors positioned at the Grid units, ESS output, and the final 120V DC Bus are used to carry out real-time monitoring. Microcontroller based Control, which uses a system model to determine the best course of action for MCU control moves and allows for proactive resource allocation, is used to accomplish the core control. A feedback control loop is added to this for instant load stabilization. Both the Change Over Relay and the ESS deployment are managed by the combined strategy, which guarantees immediate stability and maximum efficiency.

Conclusion and Future Scope:

We were able to successfully design and demonstrate a reliable LVDC microgrid power management system. In comparison to traditional techniques, the system achieves superior load stabilization and optimized power flow by utilizing an Advanced Monitoring and Control Mechanism on an affordable ESP-32 platform. A high level of dependability is ensured by the ESS's strategic integration for proactive support against voltage fluctuations. This all-inclusive control architecture offers a reliable, clever, and effective solution that is prepared for installation in contemporary DC power distribution networks.

Future Scope: The proposed monitoring and control system for the 120V DC microgrid can be further enhanced by integrating intelligent algorithms such as machine learning–based predictive control for dynamic load management and fault detection. Future developments may also include IoT connectivity for remote monitoring, real-time data logging, and adaptive energy distribution between multiple SMPS units or renewable sources. Additionally, the system can be expanded to support higher power levels, bidirectional power flow, and grid-interactive operations, making it suitable for smart grid.

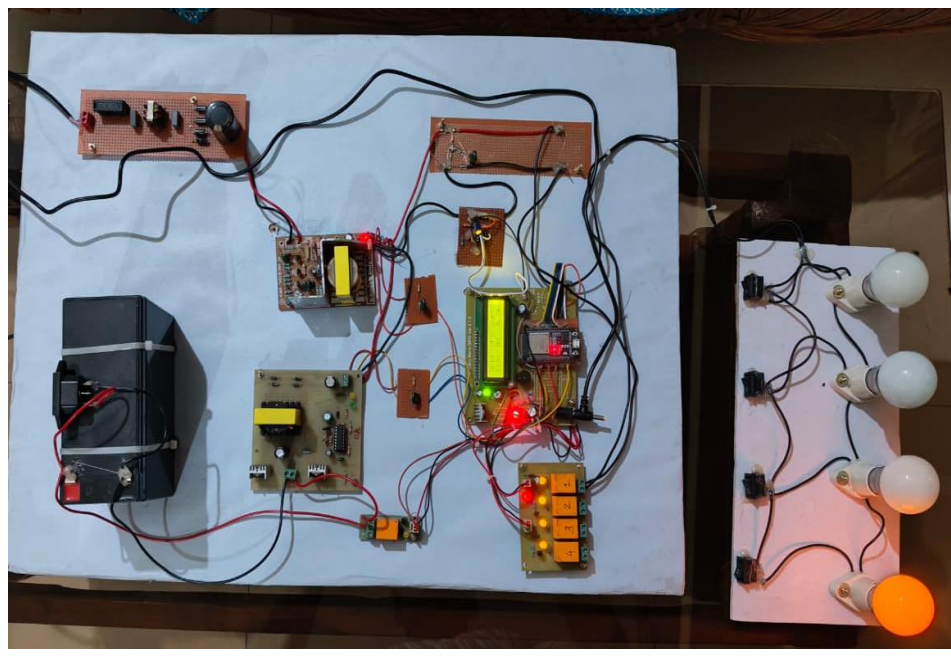
Showed Diagram:



Results: Server Record & LCD Display

• Grid is Active :				
Sr.no.	Loading	Voltage	Current	Remark
1	No load	120V	0A	System healthy
2	1 bulb	114V	0.27A	System healthy
3	2 bulbs	113V	0.55A	System healthy
4	>2 bulbs	<110V	>0.7A	Overload detected and system shutdown
• UPS Active and grid is failed :				
Sr.no.	Loading	Voltage	Current	Remark
1	No load	138V	0A	System healthy
2	1 bulb	123V	0.27A	System healthy
3	2 bulbs	122V	0.55A	System healthy
4	>2 bulbs	<118V	>0.7A	Overload detected and system shutdown

Hardware:



REFERENCE:

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	Course Matrix (CO PO Mapping Table)														
Bloom's Level		PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PSO 1	PS O2	
	L2	CO1	3	3		2							1	3	2
	L3	CO2		3		3			2				2	2	3
	L6	CO3			3		3		2		2		2		3
	L4	CO4	3			2	2				2			2	3
	L5	CO5							2		3	2	2	2	2